Year 12 Maths - Force and motion

Similar questions - Newton's second law in vector form and in lifts

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Particle P of mass 3 kg moves on a smooth horizontal plane under the action of two forces, $(4\mathbf{i} - 7\mathbf{j})$ N and $\mathbf{F}$ N. Its acceleration is $(2\mathbf{i} + \mathbf{j})$ $\mathrm{m\,s^{-2}}$.

(a) Find $\mathbf{F}$. **[3]**

(b) Find the magnitude of the acceleration and the angle it makes with the vector $\mathbf{i}$. **[3]**

(c) P starts from rest. Find its speed and its displacement from its starting point after 4 seconds. **[4]**

2 A lift of mass 340 kg carries a passenger of mass 60 kg. The lift is raised by a vertical cable, which can withstand a maximum tension of 6000 N.

(a) Find the maximum upward acceleration of the lift. **[3]**

(b) Find the magnitude of the reaction force between the passenger and the floor of the lift when the lift accelerates upwards at this maximum value. **[3]**

(c) The tension in the cable is later reduced to 3600 N. Find the acceleration of the lift (stating its direction) and the new reaction force on the passenger. **[4]**

Question 1

(a) $(4\mathbf{i}-7\mathbf{j})+\mathbf{F}=3(2\mathbf{i}+\mathbf{j})=6\mathbf{i}+3\mathbf{j}$, so $\mathbf{F}=2\mathbf{i}+10\mathbf{j}$

(b) $|\mathbf{a}|=\sqrt{4+1}=\sqrt{5}=2.24\ \mathrm{m\,s^{-2}}$. $\tan\theta=\frac{1}{2}$, so $\theta=26.6^\circ$ above $\mathbf{i}$

(c) $\mathbf{v}=\mathbf{a}t=8\mathbf{i}+4\mathbf{j}$, speed $=\sqrt{80}=8.94\ \mathrm{m\,s^{-1}}$. $\mathbf{s}=\frac{1}{2}\mathbf{a}t^2=16\mathbf{i}+8\mathbf{j}$ m

Question 2

(a) Total mass 400 kg. $6000-400g=400a$, so $a=\dfrac{6000-3920}{400}=5.2\ \mathrm{m\,s^{-2}}$

(b) For the passenger: $R-60g=60a$, so $R=60(9.8+5.2)=900$ N

(c) $3600-3920=400a$, so $a=-0.8$: $0.8\ \mathrm{m\,s^{-2}}$ downwards (the lift is accelerating downwards). $R-588=60(-0.8)$, so $R=540$ N